AI-Powered Inventory Management System

Built with n8n, Google Gemini and Google Sheets

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Repository: github.com/Star-obiero/Semester-3.1-notes-and-projects (Projects / n8n inventory management)

1. Overview

This project is a conversational inventory assistant. Instead of opening a spreadsheet, searching for a product and editing cells by hand, a user simply chats with an AI agent. The agent understands the request, looks up the current stock in a Google Sheet, and updates the sheet when stock changes. The whole system is built as a no-code/low-code workflow in n8n.

At a glance	
Workflow platform	n8n (cloud)
AI model	Google Gemini Chat Model
Data store	Google Sheets (inventory sheet)
Interface	n8n built-in chat trigger
Memory	Simple Memory (keeps conversation context)

2. The problem it solves

Small and medium businesses often track stock manually in spreadsheets. This leads to common problems:

- Time wasted searching and editing rows for simple stock checks.
- Human error when typing quantities.
- Staff who are not comfortable with spreadsheets cannot easily get stock information.
- Slow response when a customer asks whether an item is available.

A chat interface lowers the barrier: anyone who can type a sentence can check or update inventory.

3. How the workflow is built

The workflow has six nodes connected around one central AI Agent node.

Node	Type	Role
Chat trigger	Trigger	Receives the user's message and starts the workflow.

Node	Type	Role
AI Agent	Agent	Reads the message, decides what to do, and calls tools to get or change data.
Google Gemini Chat Model	Chat model	The language model that gives the agent its reasoning and natural-language ability.
Simple Memory	Memory	Stores recent messages so the agent remembers earlier parts of the conversation.
Check Inventory Stock	Tool (Google Sheets: read)	Reads rows from the inventory sheet so the agent can report stock levels.
Update Stock	Tool (Google Sheets: append or update)	Adds a new item or updates the quantity of an existing item in the sheet.

Data flow

User message → Chat trigger → AI Agent (uses Gemini + Memory) → chooses a tool (Check Inventory Stock or Update Stock) → Google Sheet → agent writes a reply back to the user.

4. Example interactions

Illustrative examples of how a user can talk to the system (replace with screenshots from your own test runs):

User says	What the agent does
How many units of canvas are left?	Calls Check Inventory Stock, finds the row, and replies with the quantity.
Add 20 units of canvas.	Calls Update Stock to increase the quantity, then confirms the new total.
Add a new item: tent poles, 50 pieces.	Calls Update Stock (append) to create a new row.
And how many did I add earlier?	Uses Simple Memory to recall the previous message in the conversation.

5. How to set it up

- Create a Google Sheet with columns such as Item, Quantity, and Reorder Level.
- In n8n, add a Chat Trigger node and connect it to an AI Agent node.
- Attach the Google Gemini Chat Model (add your Gemini API key under credentials).
- Attach Simple Memory to the agent.
- Add two Google Sheets tools: one set to read rows, one set to append or update a row. Connect your Google account and select your sheet.
- Write a short system prompt telling the agent it is an inventory assistant and when to use each tool.
- Click Open chat and test with sample questions, then export the workflow as a JSON file.

Tip: describe each tool clearly in n8n (for example, "Use this to look up the current quantity of an item") so the agent picks the right one.

6. Current limitations and next steps

Current version

- Works on one sheet and one chat interface.
- No automatic alerts: the user has to ask about stock.
- Relies on clean, consistent item names in the sheet.
- No approval step before the agent changes stock.

Planned improvements

- Low-stock alerts sent by Gmail or Telegram when quantity falls below the reorder level.
- Connect to WhatsApp or Telegram so staff can chat from their phones.
- Add a human approval step for large stock changes.
- Add a log sheet recording who changed what and when.
- Demand forecasting from sales history using Python.

7. Skills demonstrated

- Workflow automation with n8n.
- Building AI agents with tools and memory.
- Integrating a large language model (Gemini) with business data in Google Sheets.
- Designing a practical solution around a real business problem.

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